1.29 Polar Coordinates - Cartesian Equations

Worked Solutions

1 a)

$x = r\cos\theta$ and $y = r\sin\theta$

1 b)

$r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(\frac{y}{x})$

2 a)

$\sqrt{x^2 + y^2} = 3$ or $x^2 + y^2 = 9$

2 b)

$y = 5$

2 c)

$x = 2$

2 d)

$r = 4 + 4\cos\theta$

Multiply both sides by r:

$$r^2 = 4r + 4r\cos\theta$$

$$x^2 + y^2 = 4\sqrt{x^2 + y^2} + 4x$$

2 e)

$r = 2 - 2\sin\theta$

Multiply both sides by r:

$$r^2 = 2r - 2r\sin\theta$$

$$x^2 + y^2 = 2\sqrt{x^2 + y^2} - 2y$$

2 f)

$r + r\cos\theta = 4$

$$\sqrt{x^2 + y^2} + x = 4$$

2 g)

Multiply through by r:

$$r^2 = 4r\cos\theta + 2r\sin\theta$$

$$x^2 + y^2 = 4x + 2y$$

2 h)

Multiply through by r:

$$r^2 = -2r\cos\theta$$
$$x^2 + y^2 = -2x$$

2 i)

$3r + r\sin\theta = 1$

$$3\sqrt{x^2+y^2} + y = 1$$

2 j)

Multiply through by r:

$$r = \tan\theta$$
$$r = \frac{r\sin\theta}{r\cos\theta}$$
$$\sqrt{x^2+y^2} = \frac{y}{x}$$

3 a)

$r^2 = 25$

So $r = 5$

3 b)

$r^2 = 4r\cos\theta$

$r = 4\cos\theta$

3 c)

$r\cos\theta \times r\sin\theta = 9$

$$r^2 = \frac{9}{\cos\theta\sin\theta}$$
$$r = \sqrt{\frac{9}{\cos\theta\sin\theta}}$$

3 d)

$r\sin\theta = (r\cos\theta)^2 - 3r\cos\theta$

$$\sin\theta = r\cos^2\theta - 3\cos\theta$$
$$\sin\theta + 3\cos\theta = r\cos^2\theta$$
$$r = \frac{\sin\theta + 3\cos\theta}{\cos^2\theta}$$

3 e)

$\frac{(r\cos\theta)^2}{16} + \frac{(r\sin\theta)^2}{9} = 1$

$$\frac{r^2\cos^2\theta}{16} + \frac{r^2\sin^2\theta}{9} = 1$$
$$9r^2\cos^2\theta + 16r^2\sin^2\theta = 144$$
$$r^2(9\cos^2\theta + 16\sin^2\theta) = 144$$
$$r^2 = \frac{144}{9\cos^2\theta + 16\sin^2\theta}$$
$$r = \sqrt{\frac{144}{9\cos^2\theta + 16\sin^2\theta}}$$

3 f)

$(r^2)^2 = (r\cos\theta)^2 - (r\sin\theta)^2$

$$r^4 = r^2\cos^2\theta - r^2\sin^2\theta$$
$$r^2 = \cos^2\theta - \sin^2\theta$$
$$r = \sqrt{\cos^2\theta - \sin^2\theta}$$

3 g)

$(r\cos\theta)^2(r\sin\theta) = 6$

$$r^3\cos^2\theta\sin\theta = 6$$
$$r = \sqrt[3]{\frac{6}{\cos^2\theta\sin\theta}}$$

4 a)

Multiply through by r:

$$r^2 = 2r\cos\theta - 5r\sin\theta$$
$$x^2 + y^2 = 2x - 5y$$
$$x^2 + y^2 - 2x + 5y = 0$$
$$(x-1)^2 + (y+\frac{5}{2})^2 - 1 - \frac{25}{4} = 0$$
$$(x-1)^2 + (y+\frac{5}{2})^2 = \frac{29}{4}$$

which is an equation of a circle.

4 b)

$(1, -\frac{5}{2})$

4 c)

$r = \sqrt{\frac{29}{4}} = \frac{\sqrt{29}}{2}$

5)

$(r\cos\theta)^2 + (r\sin\theta)^2 + 4r\sin\theta = 12$

$$r^2\cos^2\theta + r^2\sin^2\theta + 4r\sin\theta = 12$$

$$r^2(\cos^2\theta + \sin^2\theta) + 4r\sin\theta = 12$$

$$r^2 + 4r\sin\theta = 12$$

$$(r + 2\sin\theta)^2 - 4\sin^2\theta = 12$$

$$(r + 2\sin\theta)^2 = 12 + 4\sin^2\theta$$

$$r = -2\sin\theta \pm \sqrt{12 + 4\sin^2\theta}$$

Since $r > 0$

$$r = -2\sin\theta + \sqrt{12 + 4\sin^2\theta} \quad \text{as required.}$$

6 a)

Multiply through by r:

$$r^2 = ar\cos\theta$$

$$x^2 + y^2 = ax$$

6 b)

Multiply through by r:

$$r^2 = ar\sin\theta$$

$$x^2 + y^2 = ay$$

6 c)

Multiply through by r:

$$r^2 = a(r + r\cos\theta)$$

$$x^2 + y^2 = a(\sqrt{x^2 + y^2} + x)$$

6 d)

Multiply through by r:

$$r^2 = a(r - r\sin\theta)$$

$$x^2 + y^2 = a(\sqrt{x^2 + y^2} - y)$$

7 a)

Multiply through by r:

$$r^2 = a(r\cos\theta + r\sin\theta)$$

$$x^2 + y^2 = a(x + y)$$

$$x^2 + y^2 - ax - ay = 0$$

$$(x - \frac{a}{2})^2 + (y - \frac{a}{2})^2 - 2(\frac{a}{2})^2 = 0$$

$$(x - \frac{a}{2})^2 + (y - \frac{a}{2})^2 = \frac{a^2}{2}$$

Radius is $\frac{a}{\sqrt{2}} = \frac{\sqrt{2}}{2}a$.

7 b)

Centre at $(\frac{a}{2}, \frac{a}{2})$

8 a)

Multiply through by r:

$$r^2 = a(pr\cos\theta + qr\sin\theta)$$
$$x^2 + y^2 = a(px + qy)$$
$$x^2 + y^2 - apx - aqy = 0$$
$$(x - \frac{ap}{2})^2 + (y - \frac{aq}{2})^2 - (\frac{ap}{2})^2 - (\frac{aq}{2})^2 = 0$$
$$(x - \frac{ap}{2})^2 + (y - \frac{aq}{2})^2 = \frac{a^2}{4}(p^2 + q^2)$$

8 b)

Centre at $(\frac{ap}{2}, \frac{aq}{2})$

8 c)

Radius is $\frac{a}{2}\sqrt{p^2 + q^2}$

9 a)

Centre at $(8, -6)$, radius 10

$$(x-8)^2 + (y+6)^2 = 100$$

When $x = 0$:

$$(y+6)^2 = 36 \Rightarrow y = -6 \pm 6 \text{ so } (0,0) \text{ and } (0,-12)$$

When $y = 0$:

$$(x-8)^2 = 64 \Rightarrow x = 8 \pm 8 \text{ so } (16,0) \text{ and } (0,0)$$

9 b)

$(x-8)^2 + (y+6)^2 = 100$

$$(r\cos\theta - 8)^2 + (r\sin\theta + 6)^2 = 100$$
$$r^2\cos^2\theta - 16r\cos\theta + 64 + r^2\sin^2\theta + 12r\sin\theta + 36 = 100$$
$$r^2(\cos^2\theta + \sin^2\theta) - 16r\cos\theta + 12r\sin\theta = 0$$
$$r^2 = 16r\cos\theta - 12r\sin\theta$$

Since $r \neq 0$

$$r = 16\cos\theta - 12\sin\theta$$

10 a)

Centre at $(-2, 4)$, radius $\sqrt{20}$

$$(x+2)^2 + (y-4)^2 = 20$$

When $x = 0$:

$$(y-4)^2 = 16 \Rightarrow y = 4 \pm 4 \text{ so } (0,8) \text{ and } (0,0)$$

When $y = 0$:

$$(x+2)^2 = 4 \Rightarrow x = -2 \pm 2 \text{ so } (0,0) \text{ and } (-4,0)$$

10 b)

$(x+2)^2 + (y-4)^2 = 20$

$$(r\cos\theta + 2)^2 + (r\sin\theta - 4)^2 = 20$$

$$r^2\cos^2\theta + 4r\cos\theta + 4 + r^2\sin^2\theta - 8r\sin\theta + 16 = 20$$

$$r^2(\cos^2\theta + \sin^2\theta) + 4r\cos\theta - 8r\sin\theta = 0$$

$$r^2 = 8r\sin\theta - 4r\cos\theta$$

Since $r \neq 0$

$$r = 8\sin\theta - 4\cos\theta$$

11 a)

Perpendicular bisector of $(0, 4)$ and $(0, -10)$

Midpoint: $(0, -3)$ so $y = -3$

11 b)

$y = -3$

$$r\sin\theta = -3$$

$$r = -\frac{3}{\sin\theta}$$

12 a)

$\sqrt{x^2+y^2} = \frac{1}{\sqrt{1}}$

$$x^2 + y^2 = 1$$

12 b)

$r^2 = \frac{16}{16\cos^2\theta + \sin^2\theta}$

$$1 = \frac{16}{16r^2\cos^2\theta + r^2\sin^2\theta}$$

$$1 = \frac{16}{16x^2 + y^2}$$

$$16x^2 + y^2 = 16$$

$$x^2 + \frac{y^2}{16} = 1$$

12 c)

$r^2 = \frac{400}{25\cos^2\theta + 16\sin^2\theta}$

$$1 = \frac{400}{25r^2\cos^2\theta + 16r^2\sin^2\theta}$$

$$1 = \frac{400}{25x^2 + 16y^2}$$

$$25x^2 + 16y^2 = 400$$

$$\frac{x^2}{16} + \frac{y^2}{25} = 1$$

12 d)

$r^2 = \frac{36}{9\cos^2\theta + 4\sin^2\theta}$

$$1 = \frac{36}{9r^2\cos^2\theta + 4r^2\sin^2\theta}$$

$$1 = \frac{36}{9x^2 + 4y^2}$$

$$9x^2 + 4y^2 = 36$$

$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$

13 a)

$\frac{(r\cos\theta)^2}{4} + \frac{(r\sin\theta)^2}{25} = 1$

$$25r^2\cos^2\theta + 4r^2\sin^2\theta = 100$$

$$r^2(25\cos^2\theta + 4\sin^2\theta) = 100$$

$$r^2 = \frac{100}{25\cos^2\theta + 4\sin^2\theta}$$

$$r = \frac{10}{\sqrt{25\cos^2\theta + 4\sin^2\theta}}$$

13 b)

$\frac{(r\cos\theta)^2}{16} + \frac{(r\sin\theta)^2}{9} = 1$

$$9r^2\cos^2\theta + 16r^2\sin^2\theta = 144$$

$$r^2(9\cos^2\theta + 16\sin^2\theta) = 144$$

$$r^2 = \frac{144}{9\cos^2\theta + 16\sin^2\theta}$$

$$r = \frac{12}{\sqrt{9\cos^2\theta + 16\sin^2\theta}}$$

13 c)

$\frac{9}{16}(r\cos\theta)^2 + \frac{49}{64}(r\sin\theta)^2 = 1$

$$36r^2\cos^2\theta + 49r^2\sin^2\theta = 64$$

$$r^2(36\cos^2\theta + 49\sin^2\theta) = 64$$

$$r^2 = \frac{64}{36\cos^2\theta + 49\sin^2\theta}$$

$$r = \frac{8}{\sqrt{36\cos^2\theta + 49\sin^2\theta}}$$

14 a)

$-1 \leq \sin\theta \leq 1$

So maximum of $r = \frac{6}{4-2} = \frac{6}{2} = 3$

14 b)

$-1 \leq \sin\theta \leq 1$

So minimum of $r = \frac{6}{4+2} = \frac{6}{6} = 1$

14 c)

$r(4 + 2\sin\theta) = 6$

$$4r + 2r\sin\theta = 6$$
$$4\sqrt{x^2+y^2} + 2y = 6$$
$$\sqrt{x^2+y^2} = \frac{6-2y}{4}$$
$$x^2 + y^2 = (\frac{6-2y}{4})^2$$
$$x^2 = (\frac{6-2y}{4})^2 - y^2$$
$$x^2 = \frac{36-24y+4y^2}{16} - y^2$$
$$x^2 = \frac{9}{4} - \frac{3}{2}y - \frac{3}{4}y^2$$

14 d)

$x^2 = \frac{9}{4} - \frac{3}{2}y - \frac{3}{4}y^2$

$$\frac{4}{3}x^2 = 3 - 2y - y^2$$
$$\frac{4}{3}x^2 = 3 - (y+1)^2 + 1$$
$$\frac{4}{3}x^2 = -(y+1)^2 + 4$$
$$\frac{x^2}{3} + \frac{(y+1)^2}{4} = 1$$

15 a)

$-1 \leq \sin\theta \leq 1$

So maximum of $r = \frac{16}{3-1} = \frac{16}{2} = 8$

15 b)

$-1 \leq \sin\theta \leq 1$

So minimum of $r = \frac{16}{3+1} = \frac{16}{4} = 4$

15 c)

$r(3 - \sin\theta) = 16$

$$3r - r\sin\theta = 16$$
$$3\sqrt{x^2+y^2} - y = 16$$
$$\sqrt{x^2+y^2} = \frac{16+y}{3}$$

$$x^2 + y^2 = (\frac{16+y}{3})^2$$
$$x^2 = (\frac{16+y}{3})^2 - y^2$$
$$x^2 = \frac{256+32y+y^2}{9} - y^2$$
$$x^2 = \frac{256}{9} + \frac{32}{9}y - \frac{8}{9}y^2$$

15 d)

$\frac{9}{8}x^2 = 32 + 4y - y^2$

$$\frac{9}{8}x^2 = 32 - (y-2)^2 + 4$$
$$\frac{9}{8}x^2 = -(y-2)^2 + 36$$
$$\frac{x^2}{32} + \frac{(y-2)^2}{36} = 1$$

16)

$r = \frac{k}{2+\cos\theta}$

$$2r + r\cos\theta = k$$
$$2\sqrt{x^2+y^2} + x = k$$
$$\sqrt{x^2+y^2} = \frac{k-x}{2}$$
$$x^2 + y^2 = (\frac{k-x}{2})^2$$
$$y^2 = \frac{1}{4}k^2 - \frac{1}{2}kx + \frac{1}{4}x^2 - x^2$$
$$y^2 = \frac{1}{4}k^2 - \frac{1}{2}kx - \frac{3}{4}x^2$$
$$\frac{4}{3}y^2 = \frac{1}{3}k^2 - \frac{2}{3}kx - x^2$$
$$\frac{4}{3}y^2 = \frac{1}{3}k^2 - (x + \frac{1}{3}k)^2 + \frac{1}{9}k^2$$
$$\frac{4}{3}y^2 + (x + \frac{1}{3}k)^2 = \frac{4}{9}k^2$$
$$\frac{(x+\frac{1}{3}k)^2}{\frac{4}{9}k^2} + \frac{y^2}{\frac{1}{3}k^2} = 1$$

17 a)

$r^2 \sin 2\theta = 9 \Rightarrow 2r^2 \sin\theta\cos\theta = 9$

$2xy = 9$

17 b)

$r^2 \cos 2\theta = 4 \Rightarrow r^2(\cos^2\theta - \sin^2\theta) = 4$

$$r^2\cos^2\theta - r^2\sin^2\theta = 4$$
$$x^2 - y^2 = 4$$

17 c)

$r^4 \sin 4\theta = 4 \Rightarrow 2r^4 \sin 2\theta \cos 2\theta = 4$

$$4r^4 \sin\theta \cos\theta(\cos^2\theta - \sin^2\theta) = 4$$

$$r^2 \sin\theta \cos\theta(r^2\cos^2\theta - r^2\sin^2\theta) = 1$$

$$xy(x^2 - y^2) = 1$$

18)

$r^4 \cos 4\theta = 1 \Rightarrow r^4(\cos^2 2\theta - \sin^2 2\theta) = 1$

$$r^4((\cos 2\theta)^2 - (\sin 2\theta)^2) = 1$$

$$r^4((\cos^2\theta - \sin^2\theta)^2 - (2\sin\theta\cos\theta)^2) = 1$$

$$r^4(\cos^4\theta - 2\cos^2\theta\sin^2\theta + \sin^4\theta - 4\sin^2\theta\cos^2\theta) = 1$$

$$r^4\cos^4\theta - 2r^4\cos^2\theta\sin^2\theta + r^4\sin^4\theta - 4r^4\sin^2\theta\cos^2\theta = 1$$

$$x^4 - 2x^2y^2 + y^4 - 4x^2y^2 = 1$$

$$x^4 - 6x^2y^2 + y^4 = 1$$

19 a)

$(r\cos\theta)^4 + (r\sin\theta)^4 = (r\cos\theta)^2 + (r\sin\theta)^2$

$$r^4\cos^4\theta + r^4\sin^4\theta = r^2\cos^2\theta + r^2\sin^2\theta$$

$$r^2\cos^4\theta + r^2\sin^4\theta = \cos^2\theta + \sin^2\theta$$

$$r^2\cos^4\theta + r^2\sin^4\theta = 1$$

$$r^2(\cos^4\theta + \sin^4\theta) = 1$$

Since

$$\cos^4\theta + \sin^4\theta = (\cos^2\theta + \sin^2\theta)^2 - 2\cos^2\theta\sin^2\theta$$

$$= 1 - 2\cos^2\theta\sin^2\theta$$

$$= 1 - 2(\sin\theta\cos\theta)(\sin\theta\cos\theta)$$

$$= 1 - \frac{1}{2}(2\sin\theta\cos\theta)(2\sin\theta\cos\theta)$$

$$= 1 - \frac{1}{2}\sin^2 2\theta$$

Therefore

$$r^2(\cos^4\theta + \sin^4\theta) = 1 \text{ gives } r^2 = \frac{1}{\cos^4\theta + \sin^4\theta} = \frac{1}{1 - \frac{1}{2}\sin^2 2\theta}$$

So that

$$r^2 = \frac{2}{2 - \sin^2 2\theta}$$

19 b)

Since $0 \leq \sin^2 2\theta \leq 1$

The maximum r can be is when $\sin^2 2\theta = 1$

$\sin 2\theta = 1$ has solutions $\theta = \frac{\pi}{4}$ and $\theta = \frac{5\pi}{4}$

$\sin 2\theta = -1$ has solutions $\theta = \frac{3\pi}{4}$ and $\theta = \frac{7\pi}{4}$

$$r^2 = \frac{2}{2-1} = 2$$

The maximum of r is $\sqrt{2}$

So the points are $(\sqrt{2}, \frac{\pi}{4})$, $(\sqrt{2}, \frac{3\pi}{4})$, $(\sqrt{2}, \frac{5\pi}{4})$ and $(\sqrt{2}, \frac{7\pi}{4})$

20 a)

$(r\cos\theta)^3 + (r\sin\theta)^3 = 12(r\cos\theta)(r\sin\theta)$

$$r^3\cos^3\theta + r^3\sin^3\theta = 12r^2\cos\theta\sin\theta$$

$$r\cos^3\theta + r\sin^3\theta = 12\cos\theta\sin\theta$$

$$r = \frac{12\cos\theta\sin\theta}{\cos^3\theta + \sin^3\theta}$$

20 b)

$12\cos\theta\sin\theta = 0$

So either $\cos\theta = 0$ or $\sin\theta = 0$

So that $\theta \in \{\frac{\pi}{2}, \pi\}$

20 c)

$r = \frac{12\cos(\frac{\pi}{4})\sin(\frac{\pi}{4})}{\cos^3(\frac{\pi}{4})+\sin^3(\frac{\pi}{4})} = \frac{12\times\frac{\sqrt{2}}{2}\times\frac{\sqrt{2}}{2}}{(\frac{\sqrt{2}}{2})^3+(\frac{\sqrt{2}}{2})^3} = \frac{6}{\frac{2\sqrt{2}}{8}+\frac{2\sqrt{2}}{8}} = \frac{12}{\sqrt{2}} = 6\sqrt{2}$

20 d)

The line goes through polar coordinate $(6\sqrt{2}, \frac{\pi}{4})$ and the pole.

$\therefore$ line of symmetry is $y = \tan(\frac{\pi}{4})x \Rightarrow y = x$

21 a)

$\theta = \pi$ so $r = 2 - 6\sin\pi = 2$

$(-2, 0)$

21 b)

When $\sin\theta = -1$ then $r = 2 + 6 = 8$ when $\theta = -\frac{\pi}{2}$

$(8, -\frac{\pi}{2})$

21 c)

Curve C_2 is $r = 5$

So

$$5 = 2 - 6\sin\theta$$

$$\sin\theta = \frac{1}{2} \text{ which gives } \theta = -\frac{5\pi}{6} \text{ or } \theta = -\frac{\pi}{6}$$

In Cartesian coordinates:

$$M = (5\cos(-\frac{5\pi}{6}), 5\sin(-\frac{5\pi}{6})) \text{ so } (-\frac{5\sqrt{3}}{2}, -\frac{5}{2})$$

$$N = (5\cos(-\frac{\pi}{6}), 5\sin(-\frac{5\pi}{6})) \text{ so } (\frac{5\sqrt{3}}{2}, -\frac{5}{2})$$

$$Q = (8\cos(-\frac{\pi}{2}), 8\sin(-\frac{\pi}{2})) \text{ so } (0, -8)$$

So the area of triangle MNQ is

$$\frac{1}{2} \times 2 \times \frac{5\sqrt{3}}{2} \times (8 - 2.5) = \frac{55\sqrt{3}}{4} \text{ units}^2 \text{ as required.}$$

22 a)

$3r + 2r\cos\theta = 1$

$$3\sqrt{x^2 + y^2} + 2x = 1$$
$$3\sqrt{x^2 + y^2} = 1 - 2x$$
$$9(x^2 + y^2) = (1 - 2x)^2$$
$$9x^2 + 9y^2 = 1 - 4x + 4x^2$$
$$9y^2 = 1 - 4x - 5x^2$$
$$9y^2 = (1 - 5x)(1 + x)$$

22 b)

Since the terms $9y^2$, $-5x$ and $+x$ remain in both curves, it is a translation in the x-direction.

Let $x' = x - 1$

$$9y^2 = (1 - 5(x - 1))(1 + (x - 1))$$
$$= (6 - 5x)x$$

which is curve C_2 so the transformation is a translation by $\binom{1}{0}$.

23 a)

$4r - 3r\cos\theta = 7$

$$4\sqrt{x^2 + y^2} - 3x = 7$$
$$4\sqrt{x^2 + y^2} = 7 + 3x$$
$$16(x^2 + y^2) = (7 + 3x)^2$$
$$16x^2 + 16y^2 = 9x^2 + 42x + 49$$
$$7x^2 - 42x + 16y^2 = 49$$
$$7(x^2 - 6x) + 16y^2 = 49$$
$$7((x - 3)^2 - 9) + 16y^2 = 49$$
$$7(x - 3)^2 + 16y^2 = 112$$
$$\frac{(x - 3)^2}{16} + \frac{y^2}{7} = 1$$

$\therefore a = 16$ and $b = 7$

23 b)

Translation $\binom{-3}{0}$

24)

Multiply through by r:

$$r^2 = r\cos\theta - \frac{r}{\sin\theta}$$

$$r^2 = r\cos\theta - \frac{r^2}{r\sin\theta}$$

$$x^2 + y^2 = x - \frac{x^2+y^2}{y}$$

$$x^2y + y^3 = xy - x^2 - y^2$$

So the transformation is a reflection in the line $y = x$.

25 a)

Curve C_1 in polar form:

$$x^2 + y^2 - x = \sqrt{x^2+y^2}$$

$$r^2 - r\cos\theta = r$$

$$r - \cos\theta = 1$$

$$r = 1 + \cos\theta$$

Solve simultaneously:

$$1 - \sin\theta = 1 + \cos\theta$$

$$-\sin\theta = \cos\theta$$

$$\frac{\sin\theta}{\cos\theta} = -1$$

$$\tan\theta = -1$$

$$\theta = \{\frac{3\pi}{4}, \frac{7\pi}{4}\}$$

So $r = 1 - \sin\frac{3\pi}{4} = 1 - \frac{\sqrt{2}}{2}$ and $r = 1 - \sin\frac{7\pi}{4} = 1 + \frac{\sqrt{2}}{2}$

$$(1 - \frac{\sqrt{2}}{2})\cos\frac{3\pi}{4} = (1 - \frac{\sqrt{2}}{2})(-\frac{\sqrt{2}}{2}) = \frac{1}{2} - \frac{\sqrt{2}}{2}$$

$$(1 - \frac{\sqrt{2}}{2})\sin\frac{3\pi}{4} = (1 - \frac{\sqrt{2}}{2})(\frac{\sqrt{2}}{2}) = \frac{\sqrt{2}}{2} - \frac{1}{2}$$

$$(1 + \frac{\sqrt{2}}{2})\cos\frac{7\pi}{4} = (1 + \frac{\sqrt{2}}{2})(\frac{\sqrt{2}}{2}) = \frac{\sqrt{2}}{2} + \frac{1}{2}$$

$$(1 + \frac{\sqrt{2}}{2})\sin\frac{7\pi}{4} = (1 + \frac{\sqrt{2}}{2})(-\frac{\sqrt{2}}{2}) = -\frac{\sqrt{2}}{2} - \frac{1}{2}$$

So $(\frac{1}{2} - \frac{\sqrt{2}}{2}, -\frac{1}{2} + \frac{\sqrt{2}}{2})$ and $(\frac{1}{2} + \frac{\sqrt{2}}{2}, -\frac{1}{2} - \frac{\sqrt{2}}{2})$

25 b)

$y = -x$

26)

To find N: When $\cos\theta = 1$ so $r = \frac{52}{129-100} = \frac{52}{29}$

So as Cartesian coordinates: $N(\frac{52}{29}, 0)$

To find P and Q solve simultaneously:

$$r = \frac{52}{129 - 100r^2}$$

$$r(129 - 100r^2) = 52$$

$$100r^3 - 129r - 52 = 0$$

$$(2r - 1)(5r - 4)(10r + 13) = 0$$

$$r = \frac{1}{2}, r = \frac{4}{5} \text{ or } r = -\frac{13}{10}$$

Since $r > 0$, $r = \frac{1}{2}, r = \frac{4}{5}$

When $r = \frac{1}{2}$

$$\frac{1}{4} = \cos\theta \text{ so } \theta = 1.31811607\ldots$$

When $r = \frac{4}{5}$

$$\frac{16}{25} = \cos\theta \text{ so } \theta = 0.8762980612\ldots$$

So as Cartesian coordinates: $P(\frac{1}{8}, \frac{\sqrt{15}}{8})$ and $Q(\frac{64}{125}, \frac{12\sqrt{41}}{125})$

Side NP : $\sqrt{(\frac{52}{29} - \frac{1}{8})^2 + (0 - \frac{\sqrt{15}}{8})^2} = \sqrt{\frac{10149}{58}}$

Side NQ : $\sqrt{(\frac{52}{29} - \frac{64}{125})^2 + (0 - \frac{12\sqrt{41}}{125})^2} = \frac{12\sqrt{7370}}{725}$

Side PQ : $\sqrt{(\frac{64}{125} - \frac{1}{8})^2 + (\frac{12\sqrt{41}}{125} - \frac{\sqrt{15}}{8})^2} = \sqrt{\frac{381-12\sqrt{615}}{500}}$

By the cosine rule:

$$\cos\theta = \frac{NP^2 + NQ^2 - PQ^2}{2 \times NP \times NQ} = 0.986433\ldots$$

So $\theta = 9.45°$ (3 SF).